

Quantum Computing

Syllabus: Principles of Quantum Information & Quantum Computing: Introduction to Quantum Computing, Moore's law & its end, Differences between Classical & Quantum computing. Concept of qubit and its properties. Representation of qubit by Bloch sphere. Single and Two qubits. Extension to N qubits.

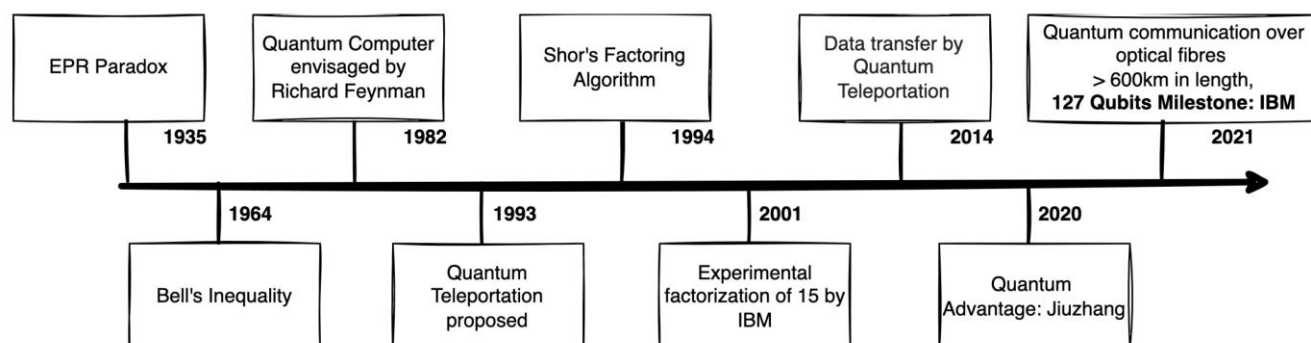
Dirac representation and matrix operations: Matrix representation of 0 and 1 States, Identity Operator I, Applying I to $|0\rangle$ and $|1\rangle$ states, Pauli Matrices and its operations on $|0\rangle$ and $|1\rangle$ states, Explanation of i) Conjugate of a matrix and ii) Transpose of a matrix. Unitary matrix U, Examples: Row and Column Matrices and their multiplication (Inner Product), Probability, and Quantum Superposition, normalization rule. Orthogonality, Orthonormality.

Quantum Gates: Single Qubit Gates: Quantum Not Gate, Pauli – X, Y & Z Gate, Hadamard Gate, Phase Gates- S & T
Multiple Qubit Gates: Controlled gate, CNOT Gate, (Discussion for 4 different input states). Representation of Swap gate, Controlled -Z gate, Toffoli gate.

Numerical problems- Identity, Unitary, Inner Product, Orthogonality, Gates: X Gates, Hadamard Gate, CNOT Gate. Relating T and S gates

Introduction to Quantum Computing:

Timeline of Quantum Computing:



Principles of Quantum Computing:

Quantum Computing is the area of study focused on developing computing methods based on the principles/laws of quantum mechanics. It is a growing field at the intersection of Physics and Computer Science. In Quantum computers, information is stored and manipulated using single isolated atoms or photons or electrons. Since atoms, molecules and photons obey the law of quantum mechanics, quantum computers perform computation and process information according to the laws of quantum mechanics. The novel features of these computers have immense potential applications in the field of cryptography. (Cryptography is the science of maintaining secrecy and security in communication)

Quantum computing is a multidisciplinary field comprising aspects of computer science, physics, and mathematics that utilizes quantum mechanics to solve complex problems faster than on classical computers.

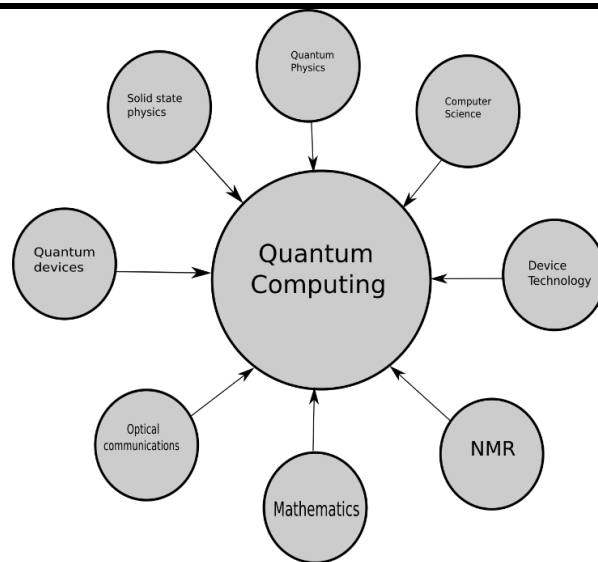
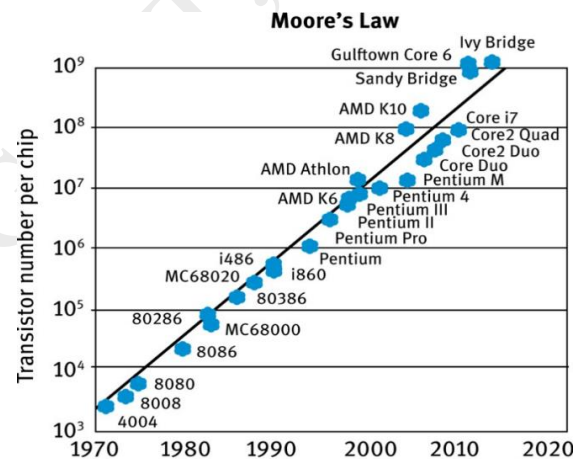


Fig: Disciplines contributing towards quantum computation

Moore's Law and its End:

In the year 1965, **Gordon Moore** observed increasing performance in the first few generations of the integrated-circuit (IC) technology. Moore predicted that, it would continue to improve at an exponential rate with the performance per unit cost increasing by one or two factor in every 18 months and so. The computer industry has followed this prediction since then. The plot shows the 50 years of Moore's law.



Statement: “The number of transistors on a microchip doubles every year”

It describes the empirical regularity that number of transistors on IC chips double approximately every two years. This advancement is important for other aspects of technological progress in computing-like processing speed.

However, reduction in the size of the transistors causes problems for current microelectronics, which results in electron tunneling. The **end** of the usage of transistors in the quantum scale could be expected as the dimension decreases further and quantum effect come into the picture. $\therefore$ Quantum Computation is the option for the further computational studies.

Concept of Qubit and its properties:

Q: Define a bit and qubit and explain the properties of qubit. (6M)

Def.- A **bit** (binary digit) is the smallest unit of data that a computer can process and store.

It is always in one of two physical states, similar to an on/off light switch. The state is represented by a single binary value, usually a **0** or **1**.

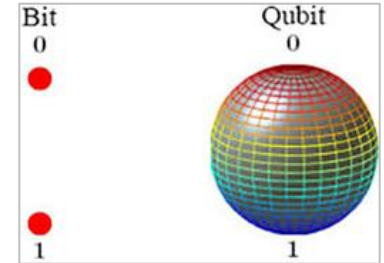
Qubit: The counter part of a classical bits in quantum computing is quantum bits, called as “qubits”.

The classical computers take, process and store everything in only two states; **0** or **1** not both. While, **quantum computers** use qubits. Qubits are similar to bits having two measurable states called 0 and 1, and also be in a superposition state of these 0 and 1 states, which can be expressed in quantum mechanical states with mathematical formula. Dirac or “bra-ket” notation is commonly used in quantum mechanics and quantum computing.

The state of a qubit is enclosed in the right half of an angled bracket, called the “**ket**”. A qubit, $|\psi\rangle$ could be in a $|0\rangle$ or $|1\rangle$ state, which is a superposition of both $|0\rangle$ or $|1\rangle$, as;

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Where α and β are the amplitude of the states.



Qubit Properties:

- **Superposition:** a qubit is a superposition of both $|0\rangle$ and $|1\rangle$. i.e., $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.
- **Entanglement** and **tunneling** are the basic properties of qubits.
(**Entanglement** - It is the state where two qubits are so strongly correlated that gaining information about one qubit will give immediate information about the other no matter how far apart these qubits are.)
(**Tunneling** - It a quantum mechanical process where qubit can penetrate through a potential barrier. The transmission through the potential barrier can be finite and relies exponentially on the barrier width and barrier height).
- **Normalization rule:** If measurements are carried out with a qubits in superposed state, then the result will be probabilistic. The total probability must be 100%. i.e., $|\alpha|^2 + |\beta|^2 = 1$
- **No cloning principle:** Due to quantum nature, the qubit changes its state during its measurements. This means, one can't copy information from qubits the way we do in present computers, as there will be no similarity between the copy and original.

Representation of Qubit by Bloch sphere:

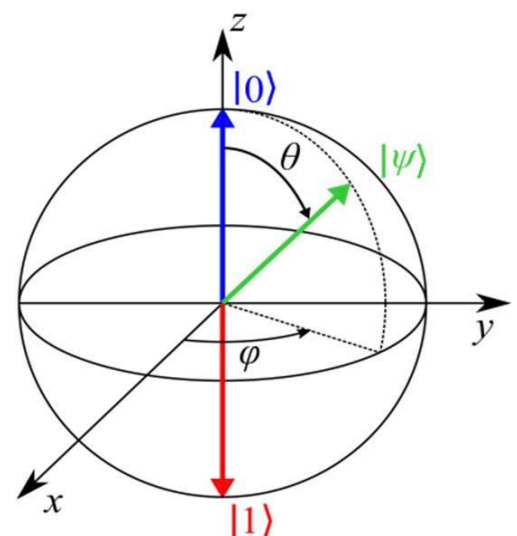
Q: Explain the representation of qubit using Bloch Sphere. (6M)

The distinct states $|0\rangle$ and $|1\rangle$ can be represented as **North** and **South** poles of a sphere of unit radius called “**Bloch Sphere**”. The state of a qubit is represented by vector in the sphere. Other locations are the superposition of $|0\rangle$ and $|1\rangle$; $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $\alpha^2 + \beta^2 = 1$. Hence, Qubit can be any point on Bloch Sphere.

The Bloch sphere allows the state of the qubit to be represented by spherical co-ordinates (θ, φ) as;

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\varphi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

Where $0 \leq \theta \leq \pi$ and $0 \leq \varphi \leq 2\pi$

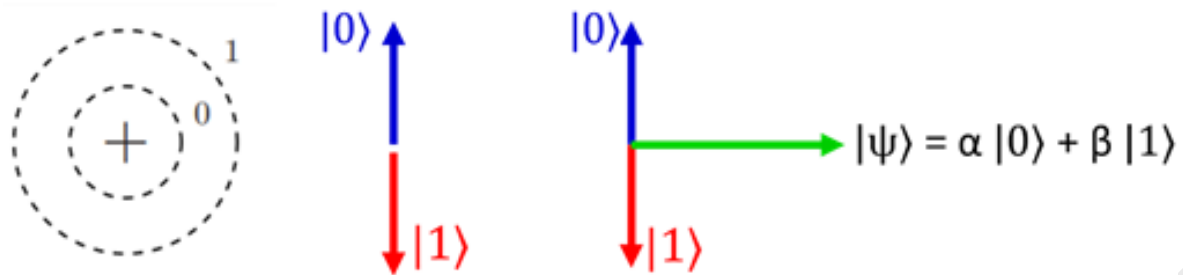


For $\varphi = 0$ and $\theta = 0$ Then, $|\psi\rangle = |0\rangle$, which is along +z axis (i.e., North Pole)

$\varphi = 0$ and $\theta = \pi$ Then, $|\psi\rangle = |1\rangle$, which is along -z axis (i.e., South Pole)

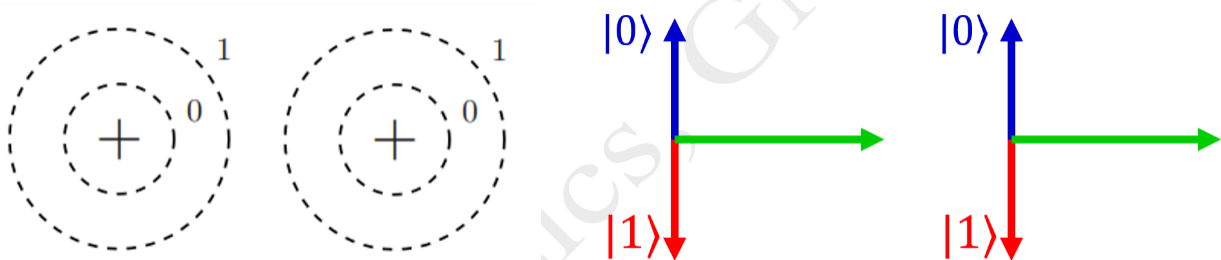
Single qubit, Two qubits and extended up to N qubits:

Single Qubit: has two computational basis states $|0\rangle$ and $|1\rangle$. The pictorial representation of the single qubit is as follows;



Let us examine a system of **single qubit**. Consider an electron in Hydrogen atom (H^1), regarded as 1-state quantum system: Since electron can be in either of the ground or excited state, classically the electron is in one of two states either 0 or 1 and represent 1 bit of classical information. By superposition principle, the quantum state of electron can be linear combination of these two states: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

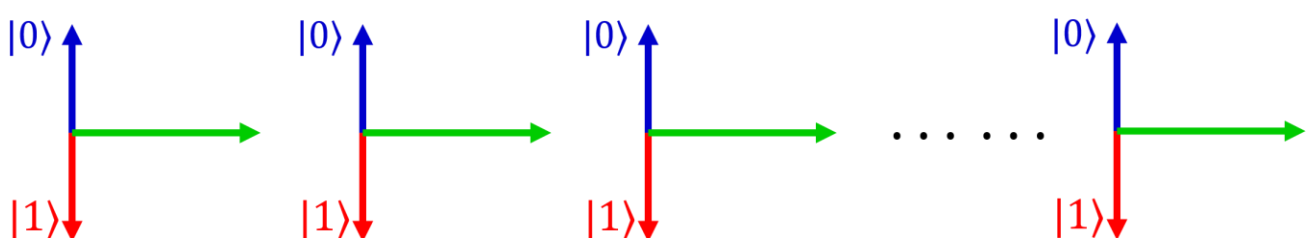
Two Qubit: has four computational basis states $|00\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$. The pictorial representation of the single qubit is as follows;



Let us examine **Two qubits**. Consider two electrons in Hydrogen atom (H^2), each electron regarded as 2-state quantum system: Since electron can be in either of the ground or excited state, classically the two electrons are in one of four states; 00, 01, 10, or 11 and represent 2 bits of classical information. By superposition principle, the quantum state of electron can be linear combination of these four states:

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

Multi-qubit system of N qubits: has 2^N computational basis states. For example, a state with 3 qubits has 2^3 computational basis states. Thus, for N qubits the computational basis states are denoted as $|00 \dots 00\rangle$, $|00 \dots 01\rangle$, $|00 \dots 10\rangle$, $|00 \dots 11\rangle \dots |11 \dots 11\rangle$. The block diagram of representation of N qubits is as follows;



Difference between Classical computer and Quantum computer:

Q: Elucidate the differences between classical and quantum computing. (6M)

Classical Computer	Quantum Computer
• Data stored with transistors are in the form 0 or 1. • Processing performed through logic gates. • Power increases in the ratio 1:1 with the number of transistors. • Have low error rate and can operate at room temperature. • Everyday processing is best handled by classical computers. • Operations are governed by Boolean algebra. • Circuit behavior is defined by classical physics. • Calculations are deterministic. It means repeating the same input results the same output.	• Data stored in qubit are either in 0 or 1 or both. • Processing performed through Quantum logic gates. • Power increases exponentially in proportion to the number of qubits. • Have high error rate and need to operate at ultra-cold conditions. • Well suited for tasks like optimization problems, data analysis and simulations. • Operations are by linear algebra and Hilbert space. • Circuit behavior is defined by quantum physics. • The calculations are probabilistic. It means there are multiple possible outputs to the same input.

Dirac Representation and Matrix Operations

Matrix representation of $|0\rangle$ and $|1\rangle$:

Q: Explain the Matrix representation of 0 and 1 States and apply identity operator I to $|0\rangle$ and $|1\rangle$ states. (6M)

In matrix representation, a **qubit** (0 or 1) is written as a two-dimensional vector, where the **states** are the components of the vector. Hence, it is useful to use matrix algebra.

i.e., $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

When writing one qubit in a superposition, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. It is useful to use matrix algebra. In matrix representation, a qubit is written as a two-dimensional vector where the **amplitudes** are the components of the vector.

i.e., $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

Identity operator (I):

“**Identity matrix** is an operator, operating on function/state, on which the function/state remains the same”.

Let the identity operator I operates on $|0\rangle$ state qubit.

$$I |0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle \quad \rightarrow I |0\rangle = |0\rangle$$

Thus, when identity operator I applied on $|0\rangle$ qubit state, remains the same after operating i.e., $|0\rangle$ qubit state.

Similarly, let the Identity I operator operates on $|1\rangle$ state qubit.

$$I |1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle \quad \rightarrow I |1\rangle = |1\rangle$$

Thus, identity operator I unchanged the $|1\rangle$ qubit state.

Pauli's Matrices and its Operation on $|0\rangle$ and $|1\rangle$ states:

Q: State the Pauli matrices and apply Pauli matrices on the states $|0\rangle$ and $|1\rangle$ (8M)

Pauli matrices are 2×2 matrices, which are very much useful in the study of quantum computation and quantum information. Usually indicated by the Greek letter sigma (σ), they used as spin operator.

Pauli matrices are given by,

$$\sigma_x = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\sigma_y = Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$\sigma_z = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Pauli matrix σ_x operates on $|0\rangle$ and $|1\rangle$ states:

$$\sigma_x |0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle \quad \rightarrow \quad \text{Pauli matrix } \sigma_x \text{ changes the } |0\rangle \text{ state to } |1\rangle$$

$$\sigma_x |1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle \quad \rightarrow \quad \text{Pauli matrix } \sigma_x \text{ changes the } |1\rangle \text{ state to } |0\rangle$$

Pauli matrix σ_y operates on $|0\rangle$ and $|1\rangle$ states:

$$\sigma_y |0\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} = i|1\rangle \quad \rightarrow \quad \text{Pauli matrix } \sigma_y \text{ changes } |0\rangle \text{ state to } i \text{ times } |1\rangle$$

$$\sigma_y |1\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -i \\ 0 \end{bmatrix} = -i|0\rangle \quad \rightarrow \quad \text{Pauli matrix } \sigma_y \text{ changes } |1\rangle \text{ state to } -i \text{ times } |0\rangle$$

Pauli matrix σ_z operates on $|0\rangle$ and $|1\rangle$ states:

$$\sigma_z |0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle \quad \rightarrow \quad \text{Pauli matrix } \sigma_z \text{ unchanged the } |0\rangle$$

$$\sigma_z |1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -|1\rangle \rightarrow \quad \text{Pauli matrix } \sigma_z \text{ changes the } |1\rangle \text{ state to } -|1\rangle$$

Explanation of Conjugate of a matrix:

It is possible to find the conjugate for a given matrix by replacing each element of the matrix with its complex conjugate. Mathematically, a conjugate matrix is a matrix, obtained by replacing the complex conjugate of all the elements of the matrix.

The conjugate of a complex number is found by just switching the sign of the imaginary part (i.e., $+i$ to $-i$ and vice versa). The complex conjugate of any real number is just a real number itself.

Ex: If A is a 2×2 complex matrix, then A^* is the complex conjugate of A ;

$$\text{i.e.,} \quad A = \begin{bmatrix} 0 & 3+i \\ -i & 1 \end{bmatrix} \quad \leftrightarrow \quad A^* = \begin{bmatrix} 0 & 3-i \\ i & 1 \end{bmatrix}$$

Explanation of Transpose of matrix:

Transpose of a matrix is defined as, “a matrix which is formed by turning all the rows of a given matrix into columns and vice-versa.”,

Ex: If A is the given matrix, then A^T is the transpose of matrix A ;

$$A = \begin{bmatrix} 0 & 3+i \\ -i & 1 \end{bmatrix} \quad \leftrightarrow \quad A^T = \begin{bmatrix} 0 & -i \\ 3+i & 1 \end{bmatrix}$$

Explanation of Conjugate Transpose of matrix:

The conjugate transpose of matrix is a matrix obtained by first transposing and then applying complex conjugate on each elements. It is denoted by the symbol $\dagger$.

Ex: If A is 2×2 matrix, then $A^\dagger$ is the conjugate transpose of matrix A .

$$A = \begin{bmatrix} 0 & 3+i \\ -i & 1 \end{bmatrix} \quad \leftrightarrow \quad A^\dagger = (A^*)^T = (A^T)^* = \begin{bmatrix} 0 & i \\ 3-i & 1 \end{bmatrix}$$

Unitary matrix:

The matrix U is said to be unitary if and only if, $UU^\dagger = U^\dagger U = I$

It is a square matrix of complex numbers. The product of the conjugate transpose of a unitary matrix, with the unitary matrix, gives an identity matrix.

$$\text{Ex: Let us consider a matrix; } U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ -i & -1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \quad \text{then} \quad U^\dagger = (U^*)^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ -i & -1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$UU^\dagger = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ -i & -1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ -i & -1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} - \frac{i^2}{2} & \frac{i}{2} - \frac{i}{2} \\ -\frac{i}{2} + \frac{i}{2} & -\frac{i^2}{2} + \frac{1}{2} \\ \frac{1}{2} + \frac{1}{2} & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$\therefore UU^\dagger = I$$

The given matrix U is unitary matrix.

Row and Column Matrices:

Row matrix: Matrix having only one row and any number columns i.e., of order $1 \times n$ is called row matrix. In quantum Physics, 'bra' vector is represented by the 'row matrix' and denoted by left half bracket ' $\langle$ '. It is actually a complex conjugate of 'ket' vector.

i.e., $\langle \Psi| = [a_{11}^* \ a_{12}^* \ a_{13}^* \dots \dots \dots a_{1n}^*]_{1 \times n}$ is the example of 'bra' vector.

Column matrix: Matrix having only one column and any number of rows i.e., of order $m \times 1$ is called column matrix. In quantum Physics, 'ket' vector is represented by 'column matrix' and denoted by right half bracket ' $| \rangle$ '.

i.e., $|\Psi\rangle = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \\ \vdots \\ \vdots \\ a_{n1} \end{bmatrix}_{n \times 1}$ is the example of 'ket' vector.

- The 'bra' is a conjugate transpose of 'ket' vector; $\{\langle \Psi| = (|\Psi\rangle^*)^T\}$ and vice-versa.
- For every 'ket' vector, there is corresponding 'bra' vector and vice-versa.

Example; $|\psi\rangle = \begin{bmatrix} 1 \\ 2+3i \\ -5i \end{bmatrix}_{3 \times 1}$ is 'ket' vector, then corresponding 'bra' is; $\langle \psi| = [1 \ 2-3i \ 5i]_{1 \times 3}$

Matrix Multiplication (Inner Product):

For a given $|\psi\rangle$ "ket" vector and $\langle \psi|$ "bra" vector, one can calculate an "inner product" as;

$$\langle \Psi| \Psi \rangle = [a_{11}^* \ a_{12}^* \ a_{13}^* \dots \dots \dots a_{1n}^*] \cdot \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \\ \vdots \\ \vdots \\ a_{n1} \end{bmatrix} = a_{11}^* a_{11} + a_{12}^* a_{21} + \dots \dots \dots + a_{n1}^* a_{1n}$$

Ex: Calculate the inner product of the given vectors; $|A\rangle = \begin{bmatrix} 1-2i \\ -3i \\ 6 \end{bmatrix}$ & $|B\rangle = \begin{bmatrix} 5 \\ 3-9i \\ -4i \end{bmatrix}$.

Solⁿ: $\langle A|B \rangle = [1+2i \ 3i \ 6] \cdot \begin{bmatrix} 5 \\ 3-9i \\ -4i \end{bmatrix}$

$$\langle A|B \rangle = 5(1+2i) + 3i(3-9i) + 6(-4i) \quad \rightarrow \quad \langle A|B \rangle = 32 - 5i$$

$$\langle B|A \rangle = [5 \ 3+9i \ 4i] \cdot \begin{bmatrix} 1-2i \\ -3i \\ 6 \end{bmatrix}$$

$$\langle B|A \rangle = 5(1-2i) + (3+9i)(-3i) + 4i(6) \quad \rightarrow \quad \langle B|A \rangle = 32 + 5i$$

Note: Inner products of any two same vectors are complex conjugate: $\langle A|B \rangle = \langle B|A \rangle^*$

Orthogonality and Orthonormality:

Q: Explain Orthogonality and Orthonormality with an example for each. (6M)

Orthogonality: The two states $|\psi\rangle$ and $|\varphi\rangle$ are said to be orthogonal, if their inner product $\langle\psi|\varphi\rangle$ is zero.

$$\langle\psi|\varphi\rangle = 0$$

Ex: Consider $|0\rangle$ and $|1\rangle$ are the two states; $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ & $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Then their inner product is, $\langle 0|1\rangle = [1 \ 0] \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = (1 \cdot 0 + 0 \cdot 1) = 0$

& $\langle 1|0\rangle = [0 \ 1] \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = (0 \cdot 1 + 1 \cdot 0) = 0$

Since, their inner product is 0, then two states $|0\rangle$ and $|1\rangle$ are said to be orthogonal.

Orthonormality: The two states $|\psi\rangle$ and $|\varphi\rangle$ are said to be orthonormal, if and only if their inner product is;

$$\langle\psi|\varphi\rangle = \delta_{\psi\varphi} = \begin{cases} 1 & \text{if } \psi = \varphi \\ 0 & \text{otherwise} \end{cases}$$

Ex: Consider the two states; $|\psi\rangle = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ & $|\varphi\rangle = |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Then their inner product is, $\langle 0|0\rangle = [1 \ 0] \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = (1 \cdot 1 + 0 \cdot 0) = 1$

$$\langle 1|1\rangle = [0 \ 1] \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = (0 \cdot 0 + 1 \cdot 1) = 1$$

$$\langle 0|1\rangle = [1 \ 0] \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = (1 \cdot 0 + 0 \cdot 1) = 0$$

$$\langle 1|0\rangle = [0 \ 1] \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = (0 \cdot 1 + 1 \cdot 0) = 0$$

Since, their inner product is 1 for $\psi = \varphi$, and 0 for $\psi \neq \varphi$. Therefore two states $|0\rangle$ and $|1\rangle$ are said to be orthonormal.

Quantum Gates

Introduction: The **classical computers** manipulate ‘bits’ using classical logic gates, such as; OR, AND, NOT, NAND, NOR, etc. Similarly, **quantum computers** manipulate ‘qubits’ using quantum logic gates. The quantum logic gates are the basic quantum circuits or mathematical entities, which are applied to qubits, the qubits state will be changes depending on which gate is applied. A quantum algorithm has to be implemented on a quantum computer using the quantum gates. After running a quantum algorithm gates, the result is retrieved by measuring the qubits state.

Q: Explain Single qubit gate and multiple qubit gate with an example for each. (8M)

Based on number of input qubits, the quantum gates are classified as; single and multiple qubit gates.

Single Qubit Gate

If the input qubits are single states such as; $|0\rangle$ and $|1\rangle$, quantum computing requires single quantum gates such as; Quantum NOT gate (X), Pauli X, Y, Z gates, Hadamard gate, and phase gates (Z, S, and T-gates).

Quantum NOT gate (X):

Q: Discuss the working of quantum not gate mentioning its matrix representation and truth-table. (6M)

In classical computer, **NOT** gate takes one input & reverse its present value. Ex: changes 0 to 1 & vice versa.

The **Quantum NOT gate (X)** is similar in that, if qubit in a state $|0\rangle$, changes to $|1\rangle$ and vice versa. When the qubit is in a superposition, then the superposition also flips.

Matrix representation: The quantum NOT gate is mathematically represented by **Pauli ‘X’** matrix;

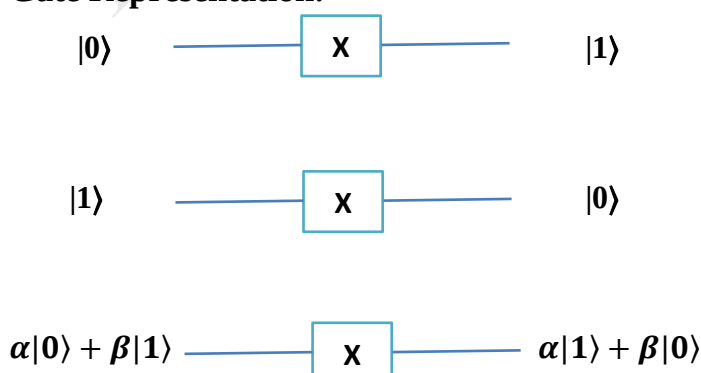
i.e.,
$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Let the Quantum NOT gate ‘X’ operates on $|0\rangle$ qubit state;
$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

Similarly, Quantum NOT gate operates on $|1\rangle$ qubit state.
$$X|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

∴ The **Quantum NOT gate (X)** reverse the given input state of the system.

Gate Representation:



Truth Table:

Inputs	Output
$ 0\rangle$	$ 1\rangle$
$ 1\rangle$	$ 0\rangle$
$\alpha 0\rangle + \beta 1\rangle$	$\alpha 1\rangle + \beta 0\rangle$

Pauli X, Y, Z gates:

Q: Discuss the working of Pauli gate mentioning its matrix representation and truth-table. (6M - any one Gate)

Pauli 'X' Gate: It is similar to the **Quantum NOT gate**.

Pauli 'Y' Gate: is a single qubit gate which operates as per the Pauli 'Y' matrix. It rotates the given input through π radians around the y-axis.

Matrix representation: The Pauli 'Y' gate is mathematically represented by Pauli ' σ_y ' matrix or 'Y';

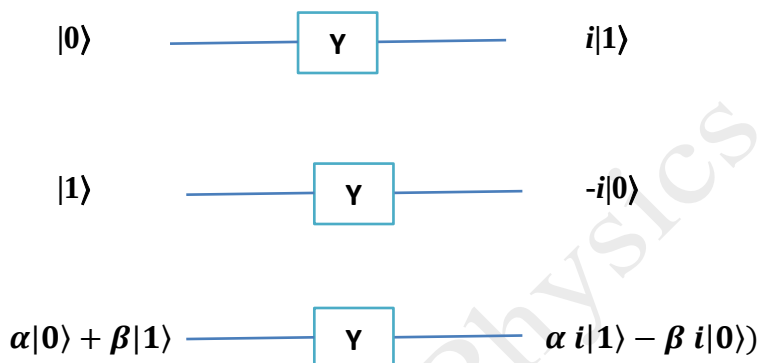
i.e.,
$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

Let the Pauli Y gate operates on $|0\rangle$ state qubit;
$$Y|0\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} = i|1\rangle$$

Similarly, Pauli Y gate operates on $|1\rangle$ state;
$$Y|1\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -i \\ 0 \end{bmatrix} = -i|0\rangle$$

$\therefore$ The Pauli 'Y' gate reverse the given input along with transformation /rotation of π radians.

Gate representation:



Truth Table:

Inputs	Output
$ 0\rangle$	$i 1\rangle$
$ 1\rangle$	$-i 0\rangle$
$\alpha 0\rangle + \beta 1\rangle$	$i(\alpha 1\rangle - \beta 0\rangle)$

Pauli Z gate: is a single qubit gate which operates as per the Pauli 'Z' matrix. Its operation takes along the z-axis.

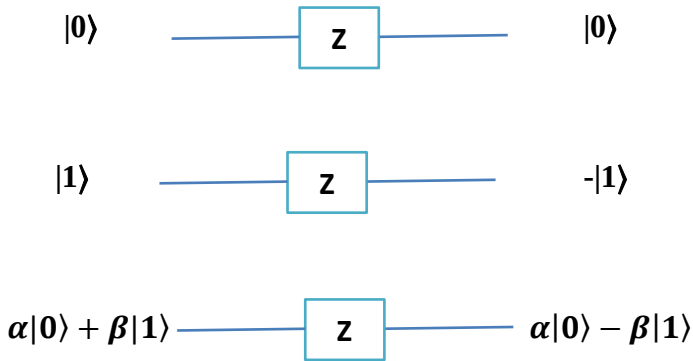
Matrix representation: The Pauli 'Z' gate is mathematically represented by Pauli ' σ_z ' matrix or 'Z';

i.e.,
$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Let the Pauli 'Z' gate operates on $|0\rangle$ state qubit;
$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

Similarly, Pauli Z gate operates on $|1\rangle$ state qubit;
$$Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -|1\rangle$$

$\therefore$ When Pauli Z gate operates on $|0\rangle$ state, which remains the same state, and operates on $|1\rangle$ state, reverse the direction of the system.

Gate Representation:**Truth Table:**

Inputs	Output
$ 0\rangle$	$ 0\rangle$
$ 1\rangle$	$- 1\rangle$
$\alpha 0\rangle + \beta 1\rangle$	$\alpha 0\rangle - \beta 1\rangle$

Hadamard Gate (H):

Hadamard gate is very important in quantum computing. If the qubit states in a definite $|0\rangle$ or $|1\rangle$ state, the Hadamard gate puts each into a superposition of $|0\rangle$ and $|1\rangle$ states.

Hadamard gate (H) satisfies the unitary condition; $HH^\dagger = H^\dagger H = I$

Matrix representation: The Hadamard gate is mathematically represented by matrix 'H';

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

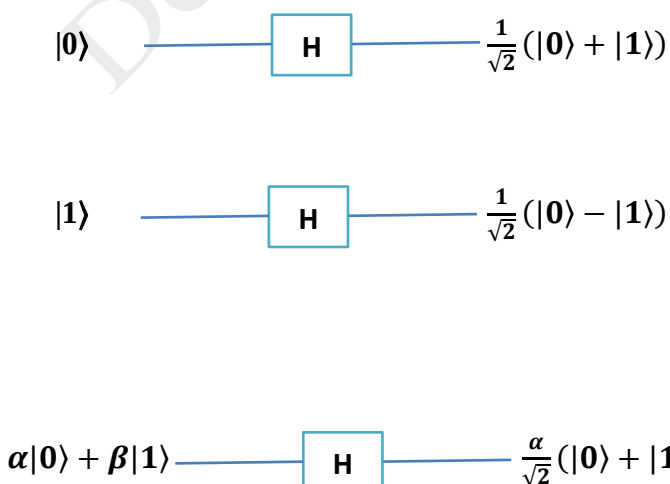
Let the Hadamard gate operates on $|0\rangle$ state qubit;

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

Similarly, let the Hadamard gate operates on $|1\rangle$ state qubit;

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$\therefore$ The Hadamard gate operates on $|0\rangle$ and $|1\rangle$ states, which puts each into superposition state.

Gate representation:**Truth Table:**

Inputs	Output
$ 0\rangle$	$\frac{1}{\sqrt{2}} (0\rangle + 1\rangle)$
$ 1\rangle$	$\frac{1}{\sqrt{2}} (0\rangle - 1\rangle)$
$\alpha 0\rangle + \beta 1\rangle$	$\frac{\alpha}{\sqrt{2}} (0\rangle + 1\rangle) + \frac{\beta}{\sqrt{2}} (0\rangle - 1\rangle)$

Phase gates:

- In phase gates, the phase angle Φ is responsible to manipulate the phase of quantum states.
- Phase is actually manipulating the qubit states as ; $|0\rangle \rightarrow |0\rangle$ and $|1\rangle \rightarrow e^{i\Phi}|1\rangle$.
- The probability of measuring a $|0\rangle$ or $|1\rangle$ unchanged after applying this gate, however it modifies only the phase of the qubit state.
- **Matrix representation** of phase gates is; $P = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\Phi} \end{bmatrix}$ where, $e^{i\Phi} = \cos\Phi + i\sin\Phi$
- On the basis of phase angle value; π , $\frac{\pi}{2}$, and $\frac{\pi}{4}$, the gates are classified as Z, S, and T, respectively.

Z Gate: It is similar to **Pauli Z gate**.

S Gate:

Q: Discuss the working of Phase gate mentioning its matrix representation and truth-table. (6M - any one Gate)

Phase gate (S) actually map the qubit states $|0\rangle \rightarrow |0\rangle$ and $|1\rangle \rightarrow i|1\rangle$. The probability of measuring a $|0\rangle$ or $|1\rangle$ unchanged after applying this gate, however, it only modifies the phase of the qubit state.

Phase gate (S) satisfies the unitary condition; $SS^\dagger = S^\dagger S = I$

Matrix representation: The S gate is mathematically represented by matrix 'S';

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

Let the S gate operates on $|0\rangle$ state qubit;

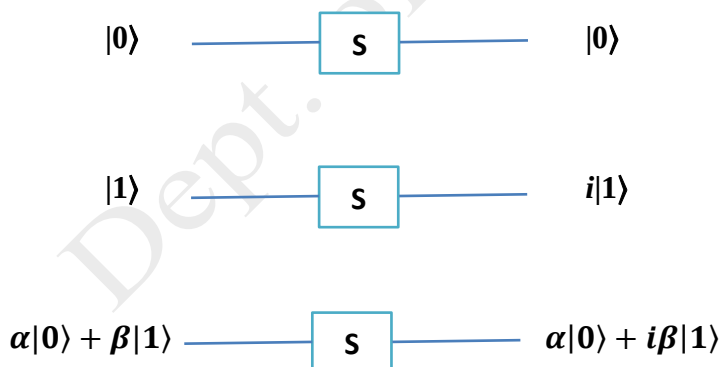
$$S|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

Similarly, let the S gate operates on $|1\rangle$ state qubit;

$$S|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} = i|1\rangle$$

∴ The S gate operates on $|0\rangle$ and $|1\rangle$ states, which remains the state and only changes their phase.

Gate Representation:



Truth Table:

Inputs	Output
$ 0\rangle$	$ 0\rangle$
$ 1\rangle$	$i 1\rangle$
$\alpha 0\rangle + \beta 1\rangle$	$\alpha 0\rangle + i\beta 1\rangle$

T Gate or $\left(\frac{\pi}{8}\right)$ gate:

Phase gate (T) actually map the qubit states $|0\rangle \rightarrow |0\rangle$ and $|1\rangle \rightarrow \frac{1+i}{2}|1\rangle$.

The probability of measuring a $|0\rangle$ or $|1\rangle$ unchanged after applying this gate, however, it only modifies the phase of the qubit state.

Matrix representation: The T gate is mathematically represented by matrix 'T';

$$T = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{bmatrix} \leftrightarrow T = e^{\frac{i\pi}{8}} \begin{bmatrix} e^{\frac{-i\pi}{8}} & 0 \\ 0 & e^{\frac{i\pi}{8}} \end{bmatrix}$$

Let the T gate operates on $|0\rangle$ state qubit;

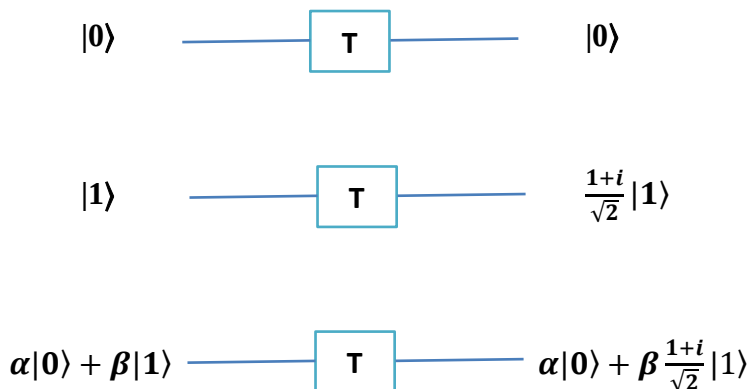
$$T|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

Similarly, let the T gate operates on $|1\rangle$ state qubit;

$$T|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1+i}{\sqrt{2}} \end{bmatrix} = \frac{1+i}{\sqrt{2}} |1\rangle$$

∴ The T gate operates on $|0\rangle$ and $|1\rangle$ states, which remains the state and only changes their phase.

Gate Representation:



Truth Table:

Inputs	Output
$ 0\rangle$	$ 0\rangle$
$ 1\rangle$	$\frac{1+i}{\sqrt{2}} 1\rangle$
$\alpha 0\rangle + \beta 1\rangle$	$\alpha 0\rangle + \beta \frac{1+i}{\sqrt{2}} 1\rangle$

Multiple Qubit Gates

The Multiple qubits are playing an important role in quantum computing. The multiple quantum gates are working on qubits containing two or more states. Among those states previous states are **Control qubits** and last one is **Target qubit**.

In multiple quantum gates one qubit state is controlled by other qubit states hence, multiple quantum gates are also called **controlled gates**.

Ex: The $|AB\rangle$ qubit is the two-state qubit, among which 'A' is **control qubit** & 'B' is **target qubit**.

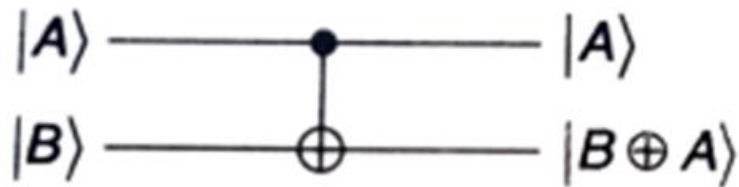
- The **target qubit** is altered if and only if the **control qubit** is $|1\rangle$, otherwise unaltered.
- The **control qubit** 'A' is unaltered during transformation.

On the basis of modes of operation, we classify the controlled gates are; Controlled NOT gate, Controlled Z-gate, Toffoli gate, Swap gate, etc.,

Controlled NOT gate (CNOT):

Q: Discuss the CNOT gate and its operation on four different input states. (6M)

- CNOT gate is a typical multi-qubit logic gate, it is similar to the conventional NOT gate, which reverse the present state of the qubit.
- In CNOT gate transformation of **Target qubit** will be controlled by **Control qubit**.
- The target state will alter the state during transformation only when control state is $|1\rangle$, otherwise unaltered.

Gate Representation;

Discussion of 4 different input states: Consider CNOT gate operating on 4 states; $|00\rangle$, $|01\rangle$, $|10\rangle$, & $|11\rangle$

1. **Operation of CNOT gate on $|00\rangle$:** Here, **control qubit** is $|0\rangle$, hence, **target qubit** doesn't alter.

$$\text{i.e., } |00\rangle \rightarrow |00\rangle$$

2. **Operation of CNOT gate on $|01\rangle$:** Here, **control qubit** is $|0\rangle$, hence, **target qubit** doesn't alter.

$$\text{i.e., } |01\rangle \rightarrow |01\rangle$$

3. **Operation of CNOT gate on $|10\rangle$:** Here, **control qubit** is $|1\rangle$, $\therefore$ **target qubit** is altered from $|0\rangle$ to $|1\rangle$

$$\text{i.e., } |10\rangle \rightarrow |11\rangle$$

4. **Operation of CNOT gate on $|11\rangle$:** Here, **control qubit** $|1\rangle$, $\therefore$ **target qubit** is altered from $|1\rangle$ to $|0\rangle$.

$$\text{i.e., } |11\rangle \rightarrow |10\rangle$$

Matrix representation:

It is mathematically represented by matrix ' U_{CN} ';

$$U_{CN} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

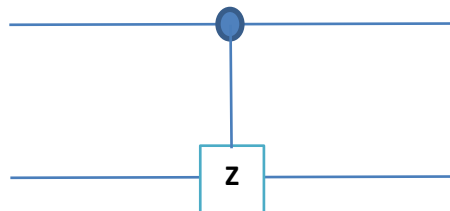
Truth Table of CNOT gate:

Inputs	Output
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 01\rangle$
$ 10\rangle$	$ 11\rangle$
$ 11\rangle$	$ 10\rangle$

Controlled Z gate (CZ):

Q: Describe the working of controlled-Z gate mentioning its matrix representation and truth-table. (6M)

- CZ gate is a typical multi-qubit logic gate, it is similar to the conventional Z gate, which transform the **Target qubit (i.e., $|B\rangle$)** as per Pauli Z operation if **control qubit** is $|1\rangle$, otherwise remains same.
- In CZ gate transformation of **Target qubit** will be controlled by **Control qubit**.

Gate Representation:**Matrix representation:**

It is mathematically represented by matrix ' U_{CZ} ';

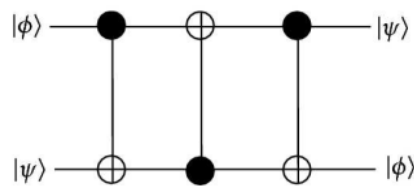
$$U_{CZ} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

Truth Table:

Inputs	Output
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 01\rangle$
$ 10\rangle$	$ 10\rangle$
$ 11\rangle$	$- 11\rangle$

Swap gate:

- **SWAP gate** is a typical two-qubit operation.
- It just swaps the state of the two qubits involved in the operation.
- It is a combined circuit of 3 CNOT gates & overall effect is; 2 input qubits are just swapped at the output.

Gate Representation:**Matrix representation:**It is mathematically represented by matrix ' U_{Swap} ';

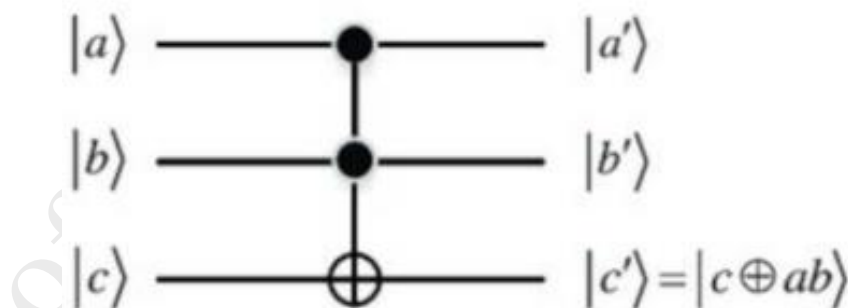
$$U_{\text{Swap}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Truth Table:

Inputs	Output
00>	00>
01>	10>
10>	01>
11>	11>

Toffoli Gate:

- **Toffoli gate**, is also known as **CCNOT** gate (Controlled-Controlled-NOT).
- It has 3 inputs, out of which **two** are **Control qubits** and **one** is **Target qubit**.
- The **Target qubit** flips only when both control qubits are |1>. The **two Control qubits** are not altered during the operation.

Gate representation:**Matrix representation:**It is mathematically represented by matrix ' U_T '

$$U_T = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Truth Table:

Inputs	Output
000>	000>
001>	001>
010>	010>
011>	011>
100>	100>
101>	101>
 110>	 111>
 111>	 110>



Solved Examples

Example 1. A linear operator 'X' operates on qubit states such that $X|0\rangle = |1\rangle$ and $X|1\rangle = |0\rangle$. Find the matrix representation of 'X'.

Solⁿ: Given $X|0\rangle = |1\rangle$

$$X|1\rangle = |0\rangle$$

$$X = ?$$

Consider X is 2×2 matrix;
$$X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}$$

From, $X|0\rangle = |1\rangle$

$$\begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} x_{11} \times 1 + x_{12} \times 0 \\ x_{21} \times 1 + x_{22} \times 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Hence, $x_{11} = 0$ and $x_{21} = 1$

Then from, $X|1\rangle = |0\rangle$

$$\begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} x_{11} \times 0 + x_{12} \times 1 \\ x_{21} \times 0 + x_{22} \times 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Hence, $x_{12} = 1$ and $x_{22} = 0$

Then X becomes;
$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Example 2. Show that given matrix $U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ i & -i \end{bmatrix}$ is unitary or not.

Solⁿ: The matrix is said to be unitary if and only if, $UU^\dagger = U^\dagger U = I$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ i & -i \end{bmatrix} \quad U^\dagger = (U^*)^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix}$$

$$UU^\dagger = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ i & -i \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} + \frac{1}{2} & \frac{-i}{2} + \frac{i}{2} \\ \frac{i}{2} - \frac{i}{2} & -\frac{i^2}{2} - \frac{i^2}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} + \frac{1}{2} & 0 \\ 0 & \frac{1}{2} + \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$U^\dagger U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ i & -i \end{bmatrix} = \begin{bmatrix} \frac{1}{2} + \frac{1}{2} & \frac{-i}{2} + \frac{i}{2} \\ \frac{i}{2} - \frac{i}{2} & -\frac{i^2}{2} - \frac{i^2}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} + \frac{1}{2} & 0 \\ 0 & \frac{1}{2} + \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$\therefore UU^\dagger = U^\dagger U = I$$

Hence, the given matrix is unitary matrix.

Example 3. Using matrix multiplication, show that on applying Hadamard gate twice to a $|0\rangle$ results in its original state.

Solⁿ: Show that $HH|0\rangle = |0\rangle$

If H is a Hadamard gate, $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$$HH = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Since we know that, $I|0\rangle = |0\rangle$

$$\therefore HH|0\rangle = |0\rangle$$

Thus Hadamard gate operate twice on any state it gives same state.

Example 4. Prove that Hadamard gate is a unitary gate.

Solⁿ: Show that $HH^\dagger = H^\dagger H = I$

If H is a Hadamard gate, $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

Then $H^\dagger = (H^*)^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$$HH^\dagger = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1+1 & 1-1 \\ 1-1 & 1+1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = I$$

$$\text{and } H^\dagger H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1+1 & 1-1 \\ 1-1 & 1+1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = I$$

$$\therefore HH^\dagger = H^\dagger H = I$$

Hence proved Hadamard gate is unitary gate.

Example 5. Show that S gate can be formed by connecting two T gates in series.

Solⁿ: Show that $T^2 = S$

If T gate is; $T = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{bmatrix}$

$$T^2 = TT = \begin{bmatrix} 1 & 0 \\ 0 & \frac{(1+i)}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \frac{(1+i)}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} = S$$

$$\therefore T^2 = S$$

Example 6. Given $A = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$, Prove that $A^\dagger = A$.

Solution : Given $A = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$

We Know That $A^\dagger = (A^*)^T$

The Conjugate of Matrix A is Given by $A^* = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$

Thus $(A^*)^T = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = A$

$\therefore A^\dagger = A$

Example 7. Find the inner product of states $|1\rangle$ and $|0\rangle$ and draw conclusions on the result.

Solution: The inner product of $|0\rangle$ and $|1\rangle$ i.e., $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \langle 0| = [1 \ 0]$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \langle 1| = [0 \ 1]$

Then, $\langle 0|1\rangle = [1 \ 0] \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = (1 \cdot 0) + (0 \cdot 1) = 0$
 $\langle 1|0\rangle = [0 \ 1] \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = (0 \cdot 1) + (1 \cdot 0) = 0$ } $\Rightarrow$ Orthogonal

Conclusion: The states $|0\rangle$ and $|1\rangle$ are orthogonal.

Example 8. Given $|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$ and $|\phi\rangle = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$ Prove that $\langle\psi|\phi\rangle = \langle\phi|\psi\rangle^*$

Solution : $\langle\phi|\psi\rangle = \begin{pmatrix} \beta_1^* & \beta_2^* \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \beta_1^* \alpha_1 + \beta_2^* \alpha_2 \dots\dots\dots (1)$

$\langle\psi|\phi\rangle = \begin{pmatrix} \alpha_1^* & \alpha_2^* \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \beta_1 \alpha_1^* + \beta_2 \alpha_2^*$

$\langle\psi|\phi\rangle^* = (\beta_1 \alpha_1^* + \beta_2 \alpha_2^*)^* = \beta_1^* \alpha_1 + \beta_2^* \alpha_2 \dots\dots\dots (2)$

Thus from (1) and (2)

$\langle\psi|\phi\rangle = \langle\phi|\psi\rangle^*$